higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

PLUMBING
NQF LEVEL 4

12 MARCH 2019

This marking guideline consists of 6 pages.

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QUESTION 1: BASIC BUILDING WORK AND BUILDING A MANHOLE

1.1
- 3 m-long straight edge
- Spirit level (2)

1.2
- Ogee (or interlock) joints
- Tongue and groove joints (2)

1.3 Benching is the smooth surface√ that slopes downwards from the sides (walls) of a manhole to the channel in which the drainpipe is. √ (2)

1.4 A half channel is a length of pipe√ that has an open section running lengthways along it. √ (2)

1.5 To allow a person to reach a drain running underground√ to clear any debris from the drain√. (2)
[10]

QUESTION 2: BELOW-GROUND DRAINAGE SYSTEMS

2.1
- Site datum
- Invert levels (2)

2.2
- Determine the required gradient for the excavations.
- Determine the invert levels of the excavation.
- Set out the excavation.
- Provide plank support (shoring to the sides of the excavation).
- Follow the occupational Health and Safety Act regulation for excavation work. (Any 4 × 1) (4)

2.3
- Water level
- Boning rods
- Sight rails
- Dumpy level
- Profile boards
- Straight edge (Any 5 × 1) (5)

2.4 A rodding eye is an opening in a drainage installation√ that allows people to get into the system. √ (2)

2.5
2.5.1 1:40
2.5.2 1:60
2.5.3 1:90
(3 × 1) (3)

2.6 To be able to clear√ blockages easily and quickly√ (2)

2.7
- To support pipes by preventing movement in the pipeline
- To support fittings and prevent them from being out of their joints by changing or heightening water pressure (2)

[20]

QUESTION 3: PLUMBING MAINTENANCE AND REPAIRS

3.1
- Remove the entire section of the pipe between two couplings and screw a new section into the existing structure
- Remove only a broken bit and replace it by fitting a new section of pipe with new coupling to hold the new section. (2)

3.2
1. Switch off the electricity supply to the geyser.
2. Unscrew the cover plate.
3. Make doubly sure that there is no electricity running to the geyser.
4. Undo the live wires connected to the thermostat. Avoid working on the neutral or the earth at this time.
5. Undo the terminal on the element. The live terminal is always indicated with a red wire.
6. Pull the thermostat from the geyser.
7. Push the new thermostat all the way into the socket designed for it. The new thermostat comes with its own live wire, so it is quick and easy to reconnect. (7)

3.3 The difference in height between the water surfaces in the reservoir and the user determines the pressure. √ The greater the difference, the higher the pressure. √ (2)

3.4
3.4.1 C
3.4.2 A
3.4.3 E
3.4.4 D
3.4.5 B

(5 × 1) (5)

3.5
- Plunger
- Drain rods
- Kinetic ram gun
- Coring and scraping
- Chemical clearing (Any 4 × 1) (4)

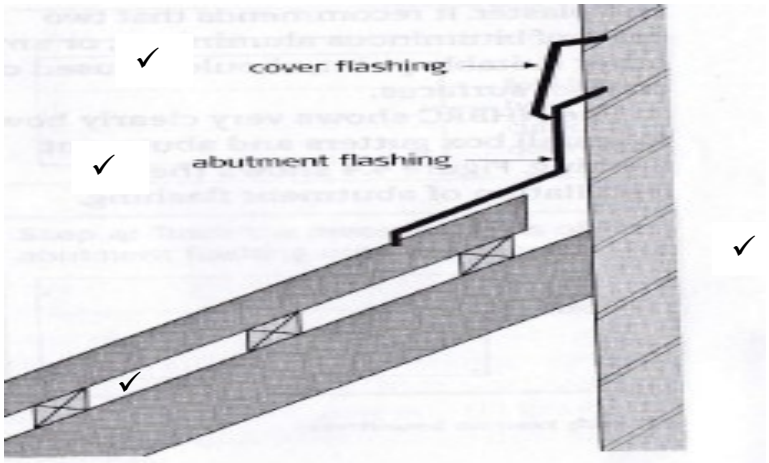
[20]

QUESTION 4: BOX GUTTERS AND CHIMNEY FLASHINGS

4.1 It is a rectangular gutter positioned inside √or against a wall which does not hang over the edge of roof. √ (2)

4.2 Flashing prevents rainwater from seeping in between the roof and the brickwork √or gutter to the inside of the building. √ (2)

4.3



(4)

4.4
- Tin snip
- Guillotine (2)

4.5
- Zinc
- Zinc alloy
- Aluminium alloy
- Copper
- Lead (Any 3x1) (3)

4.6 Soakers are pieces of metal used to water-proof √between the wall and the roof covering. √ (2)

[15]

QUESTION 5: INSTALL A SOLAR WATER HEATER

5.1 Radiant energy (1)

5.2 Solar energy is the energy given off by the sun√ in the form of heat and light. √ (2)

5.3
- Sunlight enters the solar collector box through a glass cover.
- When it reaches the black metal absorber plate inside the box, the sunlight turns to heat.
- Insulation keeps this newly created heat from escaping.
- Heat builds up and in turn heats up the water in metal tubes which are part of the collector and lie on top of the absorber plate.
- This hot water flows or is transported to a storage tank.
- An additional heater powered by electricity can be used to keep this hot water hot. This uses less electricity than heating up cold water. (6)

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5.4
- Pumped-direct systems
- Pumped-indirect systems
- Drain-back systems (3)

5.5 The back-up system will provide hot water on cloudy days√ and during periods of increased demands. √ (2)

5.6 Glazed flat-plate collectors are insulated weather-proof boxes which contain a dark absorber plate under one or more glass or plastic (polymers are acceptable) cover. √ There are flow tubes in the plate. √

Unglazed flat-plate collectors have a darker absorber plate made of metal or polymers. √ They have no cover or enclosure. √ (2 + 2) (4)

5.7 A demand water heater is a heater in which the water is heated √at the time of demand. √ (2)

5.8 The fin (or fin coating) (1)

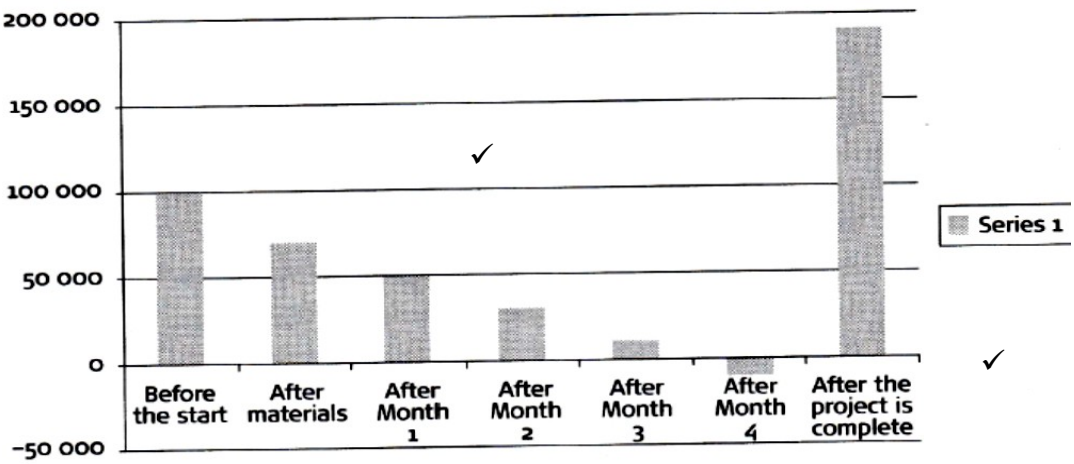
5.9 Because they lose √most of the collected energy overnight√ (2)

5.10 The position of the solar collector √relative to the movement of the sun√ (2)

[25]

QUESTION 6: BASIC BUSINESS CONCEPTS

6.1



- Before the start: R100 000
- After materials: R70 000
- After Month 1: R50 000
- After Month 2: R30 000
- After Month 3: R10 000
- After Month 4: -R10 000
- After the project is complete R190 000 (6)

6.2
- The manner in which you communicate
- Correct information that should be given to the customer
- When to contact the customer during a job (Any 2 × 1) (2)

6.3
- How many other businesses offer the same service in this area and are they good or can I do a better job?

- Are they very busy and will there be enough work for all of us or will I be able to take some work away from them?

(2)
[10]

TOTAL: 100